# AWANTIKA SRIVASTAVA


**Machine Learning Engineer | AI & Machine Learning Solution**

+91-8920482037 | sawantika81@gmail.com | LINKEDIN | Github


## PROFILE SUMMARY

Machine Learning Engineer with **2+ years of experience developing data-driven models and optimization pipelines for real-world applications**. Skilled in **Python, NumPy, SciPy, and machine learning frameworks** for predictive modeling and analytics. Experienced in building **data pipelines, model validation workflows, and production ML systems**. Strong foundation in **statistical modeling, algorithm design, and numerical optimization techniques** to solve complex decision-making problems. Proven ability to translate business objectives into **scalable ML and optimization models**.

## CORE TECHNICAL SKILLS

- **Programming Languages:** Python, Numpy, Pandas, SciPy,Scikit-learn, C++, SQL.
- **Statistics & Mathematics:** EDA, Statistical Modeling, Descriptive Statistics, Hypothesis Testing, Probability, Sampling, Scenario Analysis.
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning.
- **Deep Learning & AI:** CNN, RNN, LSTM, BERT, Transfer Learning, Model Fine-tuning, Time-Series forecasting, ARIMA, SRIMA, TensorFlow, Keras, PyTorch
- **Optimization & Operations Research:** Linear Programming, Mixed Integer Programming, Convex Optimization, Constraint Optimization, Scenario Analysis
- **Computer Vision:** Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Video Analytics, OpenCV
- **NLP**: Text preprocessing, Tokenization, chunkings, Sentiment Analysis, Topic Modeling (LSA, LDA), Transformer-based models.
- **Development & MLOps:** Algorithm Design, Numerical Methods, Git, Docker, REST APIs, Model Monitoring, CI/CD
- **Platforms:** Linux, Jupyter Notebook, AWS EC2

## EXPERIENCE

**Machine Learning Engineer | PPS International Pvt. Ltd.** **January 2024-Present**

- Developed **machine learning models and predictive analytics solutions using Python** for real-world data applications.
- Built scalable **data pipelines and data processing workflows** to prepare datasets for model training and evaluation.
- Implemented **optimization strategies and numerical techniques** to improve model performance and decision-making efficiency.
- Designed **feature engineering workflows** to extract meaningful signals from structured datasets.
- Conducted **model validation and performance monitoring** to ensure reliability of deployed ML systems.
- Performed **scenario analysis and sensitivity analysis** to evaluate model behavior under varying business conditions.
- Developed **REST APIs for model inference** and integrated ML models with production systems.
- Collaborated with cross-functional teams to translate business problems into **mathematical models and predictive solutions**.

## PROJECTS

**Railway Driver Assistance System (RDAS) | Enterprise ML Project**

- **Designed and deployed a** real-time computer vision–based ML system **to detect unsafe driver behaviors from continuous video streams**.
- **Trained** and optimized **CNN-based object detection models (SSD MobileNet architecture)** to perform real-time inference on video data.
- Implemented **end-to-end ML pipelines** covering data ingestion, preprocessing, model training, evaluation, and production inference.
- Achieved **20–25 FPS real-time processing** with **<150 ms inference latency** by optimizing models for deployment.
- Built and integrated a **Flask-based web interface** to visualize detections and automatically recorded **30-second event clips**, reducing manual review effort.

**Chatbot Using LLM & RAG | Applied ML Project**

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

**YouTube Comments Sentiment Analyzer | link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/**

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

## CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

## EDUCATION

**IMS Engineering College, Ghaziabad** **September - 2020**

**Bachelor of Technology (Electrical and electronics engineering)**